



ENTERPRISE ETHEREUM ALLIANCE
Privacy Working Group

State of Privacy on Ethereum for Enterprise

Mapping Financial Institution Requirements to Ethereum Solutions

March 2026

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Opening Statement

The Principle Of Collaborative Neutrality

This report represents a collective effort by the Enterprise Ethereum Alliance (EEA) Privacy Working Group to map the current landscape of privacy within the Ethereum ecosystem. From its inception, this report has been guided by neutrality as a core design principle. Our aim is to provide an objective, technical, and strategic framework for understanding how privacy is being integrated into enterprise-grade blockchain solutions.

The scope of this report is intentionally focused on the privacy solutions developed and maintained by EEA member organizations. By concentrating on the innovations within our ecosystem, we can ensure that solution characteristics are supported by live pilots, peer-reviewed empirical papers or enterprise deployments. While we recognize the broader ecosystem of privacy research, deep-dives into non-member solutions fall outside the primary scope of this mandate. However, in the interest of authenticity and inclusivity, a comprehensive taxonomy of non-member solutions and academic primitives can be found in the 'Additional Privacy Solutions' section at the conclusion of this paper.

Executive Summary

Financial institutions are increasingly deploying tokenized assets on Ethereum. However, on public blockchains, all transaction data is visible by default. Transaction amounts, counterparty identities, and business relationships are exposed on infrastructure that anyone can monitor.

Privacy and confidentiality are the missing pieces for enterprise adoption.

To address this, the Enterprise Ethereum Alliance, the member-driven organization connecting Fortune 500 companies, financial institutions, and technology providers with the Ethereum ecosystem, convened this Privacy Working Group to map the current landscape. Seven EEA member organizations contributed their solutions, creating the first comprehensive enterprise-focused view of privacy capabilities on Ethereum.

This paper is a practical guide for enterprises to evaluate privacy solutions in order to utilize them for their own application of tokenized assets. It is designed for CIOs, compliance officers, and digital asset leads who need to evaluate options and ask the right questions.

Privacy has various applications and implementations. Different solutions protect different aspects of operations. No single solution covers all requirements. This paper provides the framework to understand what exists, how each approach works, and what implementation requires, so institutions can make informed decisions for their specific needs.

Research Methodology

Rigorous Cross-Institutional Analysis

The findings within this paper were developed within the EEA Privacy Working Group over a dedicated three-month intensive study period. The methodology relied on a synthesis of technical documentation, stress-testing results, and deployment post-mortems from the world's leading financial and cryptographic institutions.

The working group was led by a steering committee comprising:

- Applied Blockchain and their Silent Data team, Consensys and Linea, COTI, EY and the Nightfall teams, Kaleido, Polygon as well as ZKsync and Matter labs.

Our process involved cross-review of privacy primitives, and the consolidation of proprietary enterprise requirements into a unified set of standards for Ethereum-based privacy.

Enterprise Privacy Problem

The Barriers to Public Blockchain

For enterprises, the transparency that defines public blockchains is a "double-edged sword." While it provides immutable trust, it simultaneously creates a critical confidentiality deficit. The primary issues include:

Exposure of Business Logic:

On a public ledger, smart contract interactions can reveal sensitive business logic, trade secrets, and proprietary algorithms to competitors.

Lack of Financial Discretion:

Publicly visible transaction amounts and wallet balances are incompatible with corporate treasury requirements and high-stakes financial maneuvers.

Regulatory Non-Compliance:

Regulations such as GDPR and MiCA require strict controls over data access and the "right to be forgotten," which are fundamentally at odds with a permanent, public, transparent ledger.

Impact on Financial Strategy:

Visibility into transaction flows allows for front-running and "MEV" (Maximal Extractable Value) exploitation, which erodes the profitability of institutional trading strategies and introduces unacceptable threat actors.

Institutional Trust Requirements

Institutions require an infallible level of trust in their financial products and instruments. The option for confidentiality is a requisite to preserve this trust in the process of tokenized assets.

Without robust privacy, the public blockchain remains a "read-only" experiment for the world's largest enterprises.

Privacy Addressable Market

Unlocking the Value of On-Chain Privacy

The transition of institutional assets to the blockchain represents a multi-trillion-dollar opportunity. Privacy is not just a feature, it is the key that unlocks these sectors:

Global Finance (TradFi): The total addressable market for private decentralized finance (DeFi) includes the \$100T+ global bond market and the \$600T derivatives market, both of which require strict transaction privacy.

Why Global Finance (TradFi) Benefit from On-Chain Privacy: Privacy is the essential bridge that allows the global bond market and derivatives market to move on-chain without exposing sensitive positions, trade strategies, or client data - enabling institutions to capture DeFi efficiency while meeting strict regulatory, compliance and confidentiality requirements.

Supply Chain & Logistics: Protecting the identity of suppliers and the pricing of raw materials is essential for maintaining competitive advantages in global trade.

Why Supply Chain & Logistics Benefit from On-Chain Privacy: Tokenization unlocks real-time visibility, transparent & auditable provenance in global supply chains, but only privacy protects proprietary supplier identities, pricing negotiations, and contract terms, and other forms of sensitive data

Healthcare & Data Sovereignty: The secure, private exchange of patient records and genomic data requires a "Privacy-First" architecture to meet global compliance standards.

Why Healthcare & Data Sovereignty Benefit from On-Chain Privacy: Privacy-first tokenization enables secure, interoperable exchange of patient records, genomic data, and clinical trial results across borders and institutions - satisfying GDPR, HIPAA, and emerging global standards while giving individuals true sovereignty over their most sensitive health information.

By solving the privacy paradox, we enable the migration of high-value, sensitive assets from siloed legacy databases to a unified, global Ethereum infrastructure.

Taxonomy of Privacy Approaches in the Ethereum Ecosystem

The Technical Primitives of the Ethereum Ecosystem

To navigate the privacy landscape, we categorize solutions into five primary technologies:

Fully Homomorphic Encryption (FHE):

An emerging field that allows data to remain encrypted even while it is being processed, ensuring that sensitive information is never "unlocked" during computation.

Garbled Circuits (GC):

Distributes computation across decentralized software, enabling any one individual or multiple parties to collectively compute over data while it remains encrypted; this ensures that sensitive information is never "decrypted".

Multi-Party Computation (MPC):

Distributes a computation across multiple parties such that no single party can see the entire data set, preventing any single point of failure or data leak.

Trusted Execution Environments (TEEs):

Uses a hardware-secured "enclave" within a processor to run computations in complete isolation, ensuring that sensitive data cannot be accessed or tampered with — even by the host system or platform operator.

Zero-Knowledge Proofs (ZKPs):

A way for one party to prove to another that a statement is true without revealing any information beyond the validity of the statement itself (e.g., "I have enough funds," without showing the balance).

Privacy Groups:

A closed group of participants executing a shared business logic and maintaining a privately replicated ledger, while maintaining a state transition history on the public ledger with commitments to enforce consistency among the private ledgers.

Privacy-Related Ethereum Standards

In addition to cryptographic primitives, the Ethereum ecosystem has developed specific standards for privacy-preserving transactions:

ERC-5564 (Stealth Addresses)

Enables senders to generate private addresses for recipients without prior interaction. Recipients use viewing keys to identify incoming transfers without revealing their identity to observers.

ERC-6538 (Stealth Meta-Address Registry)

Provides a registry for stealth meta-addresses, enabling senders to discover recipients and generate stealth addresses without prior direct interaction. Complements ERC-5564 by providing the lookup mechanism required for practical deployment.

ERC-7984 (Confidential Fungible Token) [Draft]

Defines an interface for tokens with confidential balances and transfer amounts. Technology-agnostic, supporting FHE, ZK proofs, TEEs, or other privacy mechanisms.

These standards do not provide end-to-end privacy by themselves, but define interoperable building blocks upon which enterprise privacy solutions can be constructed.

The Ethereum Foundation's Privacy Roadmap

Ethereum's base protocol was designed for transparency. Every transaction, balance, and contract interaction is publicly visible by default. While this transparency enables trustless verification, it creates fundamental barriers for institutional adoption, barriers that the Ethereum Foundation has formally committed to addressing.

What Is Coming? Three Tracks of the EF Privacy Roadmap

The Ethereum Foundation's PSE team organizes its privacy work into three tracks that map directly to the layers of the Ethereum node architecture:

1. Private Proving

- a. covers the application layer: enabling users to prove facts about themselves (identity, membership, eligibility) without revealing underlying data. Projects like **Semaphore** (anonymous group membership proofs), **MACI** (collusion-resistant private voting), **zkPDF** (proving on signed PDF) and **zkEmail** (email-based attestations without revealing message content) are already in production use and provide building blocks for enterprise identity and compliance workflows.

2. Private Writes

- a. focuses on enabling confidential actions on Ethereum mainnet. PSE is developing the infrastructure for private transfers, including precompiles, reference implementations, tooling on SDK integrations for privacy-first wallets including private account recovery.

3. Private Reads

- a. tackles the most overlooked privacy gap today: metadata leakage from reading state. Every time a wallet checks a balance, queries transaction history, or looks up a contract, the RPC provider learns exactly which addresses and assets the user cares about. The EF is developing Private Information Retrieval (PIR), a cryptographic technique that allows a client to query blockchain state without the server learning what was requested, alongside browser-based network level privacy integration. The target is for major RPC providers, wallets, and block explorers to offer privacy alternatives within 12 months.

The Ethereum Foundation's goal is to deliver full-stack privacy across every layer of the protocol, from how users query data, to how transactions are submitted and proven, establishing the neutral, open infrastructure that the entire industry can build on to make privacy the default rather than the exception.

The Trust Models Of Enterprise Privacy Solutions

Trust, Verification, and Risk in Enterprise Privacy Solutions

Privacy technologies in the Ethereum ecosystem rely on different trust assumptions that directly affect enterprise risk, compliance posture, and operational assurance. When selecting a privacy solution, enterprises should evaluate who or what must be trusted, under what conditions, and how that trust can be reduced, distributed, or verified.

This section outlines the primary trust models across Zero-Knowledge (ZK), Secure Multi-Party Computation (MPC), Garbled Circuits (GC), Privacy Groups (Notary-based privacy), Trusted Execution Environments (TEE), Fully Homomorphic Encryption (FHE), and notary based privacy as used within the Ethereum ecosystem and advanced by members of the Enterprise Ethereum Alliance.

Cryptographic Trust (ZK, MPC, Garbled Circuits)

Primary assumption:

Trust is placed in publicly verifiable cryptography, not in any single operator, participant, or hardware vendor.

- **Zero-Knowledge (ZK):**

Trust is **minimized through** rigorous cryptographic, algebraic, and geometric principles. **Proofs are mathematically verifiable by anyone, removing dependence on third-party execution. In practice, security still depends on implementation integrity and secure parameter lifecycle management. While zero-knowledge proofs offer mathematical verification, their practical security is anchored in the integrity of the initial setup. For the enterprise, this means the 'trustless' nature of the system is ultimately predicated on the verifiable provenance of the underlying implementation and the absolute destruction of any sensitive parameters generated during its inception.**

Failure mode:

Circuit bugs or missing constraints can silently produce valid-looking proofs for incorrect states. These failures occur at the proof level and remain undetectable by standard verification without formal auditing

- **MPC & Garbled Circuits:**

Trust is placed in **the protocol design and mathematical construction of the circuit** and the cryptographic primitives (e.g., oblivious transfer, secret sharing) used to distribute computation. Correctness depends on protocol logic that prevents any single party from altering outputs or learning unauthorized intermediate states.

Failure mode:

Malicious inputs or implementation flaws can trigger incremental leakage through protocol interactions, enabling silent extraction of private data over time.

Failure mode:

- **Zero-Knowledge (ZK):**

The complexity of circuit design means subtle bugs or missing constraints can allow a system to generate proofs that are technically valid but represent incorrect or unintended computations. Because these failures occur at the proof level, they are often silent and can bypass standard verification checks, especially in automated systems like zk-rollups. For enterprises, this creates a risk of undetected data integrity issues or invalid state transitions, making rigorous auditing, testing, and formal verification essential.

- **MPC, Garbled Circuits:**

If the MPC software isn't hardened against malicious inputs, a rogue participant could send "trick" data to the group. By watching how the system reacts or fails, they could guess pieces of private data, one bit at a time.

The complexity of MPC circuit design and protocol coordination creates a risk that subtle implementation errors or incorrect assumptions can compromise privacy or correctness, even when the underlying cryptography is sound..

Hardware-Anchored Trust (TEE)

Primary assumption:

Trust is placed in hardware security guarantees provided by manufacturers and their associated firmware, microcode, and supply chains.

- **Trusted Execution Environments (TEE):**

Confidentiality and integrity rely on secure enclaves. Enterprises must trust both the hardware **vendor and the enclave's remote attestation mechanism**.

Failure mode:

Attestation authority compromises allows "malicious enclaves", running modified or unauthorized software, to falsely masquerade as valid, secure execution environments, effectively **bypassing the privacy guarantees of the network**. If **attestation authorities or signing keys** are compromised, malicious enclaves can run modified or unauthorized code while still appearing valid.

Leaked encryption keys for the enclave due to manufacturing process deficiency can cause the confidential data sent to the enclave to be leaked. **Manufacturing defects or weak key-management processes** can leak enclave encryption keys, exposing data processed inside the enclave.

Intel SGX fails where "side-channel" tricks eavesdrop on the processor's physical behavior, such as its power use or speed, to guess the secrets hidden inside.

Side-channel attacks (e.g., power, timing, or cache analysis) can infer enclave secrets without directly breaking enclave isolation.

Organizational Honesty (FHE Co-Processors, MPC, Notary based privacy)

Primary assumption:

Trust is placed in the honesty of organizations operating **core network components**. The **required honesty** threshold may be a majority (FHE Co-Processors;notary groups), or at least one honest party (MPC).

- **FHE with decentralized Co-Processors:**

Correct and honest FHE computations are guaranteed by the assumption that a majority of co-processor deployers and operators run correct and honest implementations.

Correct and secure FHE execution assumes a sufficient fraction of co-processor operators run honest and correct implementations.

Failure mode:

Collusion at or above the trust threshold enables full reconstruction of protected data; failure is catastrophic rather than partial.

- **MPC (Multi-Party Computation):**

At least one honest party is required to protect each participant's private inputs and ensures no party can learn more than they're supposed to from the computation. **Privacy is preserved as long as at least one participant remains honest**

Failure mode:

If all participants are compromised, the “at least one honest party” guarantee collapses and private inputs can be fully reconstructed.

- **Notary-base privacy:**

Confidentiality and integrity rely on either a designated party or a group acting as the “notary”, to validate transactions and attest approval. which inspects all transactions and signs a certificate to indicate approval.

Failure mode:

If enough notary members to meet the required threshold are compromised or colluded, they can “rubber-stamp” fraudulent transactions, enabling privacy-rule bypass or double-spending through valid-looking certificates for invalid actions.

Failure mode:

- **FHE with decentralized Co-Processors:**

Collusion among the threshold decryption parties results in the entire FHE-protected states to be leaked.

- **MPC:**

If an adversary successfully corrupts or colludes with every single participant in a specific computation, the "at least one honest party" protection vanishes, and the private data is exposed.

- **Notary base privacy:**

If the required threshold of a notary group is compromised or colluded, they can "rubber stamp" fraudulent transactions. This allows an attacker to bypass privacy rules or double-spend assets by obtaining a valid-looking certificate for an invalid action.

When evaluating privacy solutions on Ethereum, enterprises should assess:

1. **Who must be trusted?** (Vendors, network participants, hardware manufacturers, cryptography)
2. **What happens if that trust fails?** (Data leakage, incorrect computation, loss)
3. **How is trust minimized or distributed?** (Thresholds, verification, transparency)
4. **Can trust be audited or independently verified?** (Proofs, attestations, open standards)

Privacy Coverage Matrix - (March 2026 Snapshot)

About This Assessment

- **Background:** This section provides a maturity assessment of privacy solutions for enterprise Ethereum. Rather than binary capability indicators, we use maturity levels to reflect the current development stage.

- **Methodology:** Classifications are based on publicly available information, including announcements, documentation, and demo materials. Vendors are invited to review and provide updates.
- **Context:** Privacy technology for enterprise blockchain is an emerging field. Solutions at various maturity stages reflect the current state of the industry. **This matrix presents the status in March 2026.** The pace at which the technology is evolving is very fast and the reader should take this into consideration.

Maturity Levels

● Prototype	Working demonstration or testnet available
● Pilot	Deployment with one or more partners
● Early Access	Available to customers on a limited basis
● Available	Generally available with production deployments

Vendor Evidence & References

- **Prividium (ZKsync) — ● Available**
 - Project Dama 2 (Singapore MAS) with Deutsche Bank • Cari Network Tokenized Deposits for US Regional Banks • Fireblocks and BitGo integrations • \$2.2B TVL in RWAs
 - Docs: zksync.io/prividium
- **COTI — ● Available**
 - V2 Mainnet launched March 2025 • StaTwig VaccineLedger (10M tx) • Central Bank Of Europe Pioneer Partner Pilot • Privex Private Exchange (\$25bn cumulative volume)
 - Mainnet: mainnet.cotiscan.io | Docs: docs.coti.io
- **Nightfall (EY) — ● Early Access**
 - Nightfall_4 released April 2025 • Open source on GitHub • Integrated into EY OpsChain • Live on StarkNet
 - GitHub: github.com/EYBlockchain/nightfall_4_CE | Platform: blockchain.ey.com
- **Silent Data (Applied Blockchain) — ● Available**
 - Mainnet launch July 2025 • Aberdeen Money Market Fund (MMF) by Archax, Privacy-Enabled Live in Production • CRYOPDP (DHL Healthcare and Life Sciences) in production • Bank of England Digital Pound Lab Phase 1
 - Demo: youtube.com/watch?v=k62bZg898jY | BoE Lab: bankofengland.co.uk/the-digital-pound/lab
- **Paladin (LFDT, Kaleido) — ● Available**
 - Bank of Indonesia Digital Ruphia • HKMA eHKD • Bank of Brazil Drex
 - BoI Digital Ruphia: [report](#) | HKMA eHKD: [report](#)
- **Polygon CDK — Available**
 - Polygon CDK • Enterprise privacy rollup • Technical documentation available

- **Linea Enterprise (Consensys) — Prototype**

- Consensys Layer 2 solution • Enterprise focus announced • SWIFT collaboration reference - private testnet.

Website: consensys.io

Scenario 1: Regulatory Credential Verification

- *Scenario 1: Enterprise A wants to verify that Counterparty B meets regulatory criteria (jurisdiction, sanctions, accreditation) before executing on a shared settlement network, without accessing B's underlying identity documents.*

In regulated enterprise contexts, credential revocability and freshness are also relevant considerations. A revoked or stale credential should not pass verification.

Vendor	Trust	Regulatory Credential Check	Permissioned Query Rights	Demo
COTI	GC	 Available	 Available	Sandbox Environment
Linea	ZK	 Proto	 Proto	Doc
Nightfall	ZK	 Pilot	 Pilot	GitHub
Paladin	ZK/Notary	 Available	 Available	Docs
Prividium	ZK	 Available	 Available	Doc
Polygon	ZK	 Available	 Available	Docs
Silent Data	TEE	 Available	 Available	GitHub

Scenario 2: Private Salary Payment

- *Scenario 2: Send a salary payment in stablecoins with transaction privacy.*

Vendor	Trust	Tx Amt	Address	Balance	Regulatory Access	Demo
COTI	GC	 Pilot	 Pilot	 Available	 Available	Private Tx Demo
Linea	ZK	 Proto	 Proto	 Proto	 Proto	Docs
Nightfall	ZK	 Early Access	 Early Access	 Early Access	 Available	GitHub
Paladin	ZK/Notary	 Available	 Available	 Available	 Available	Docs
Prividium	ZK	 Available	 Available	 Available	 Available	Video
Polygon	ZK	 Available	 Available	 Available	 Available	Docs

Silent Data	TEE	 Available	 Available	 Available	 Available	Video
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Scenario 3: Confidential Trading Strategy

- Scenario 3: Hedge fund A is executing its proprietary quantitative trading strategy via a confidential smart contract on a blockchain platform

In institutional trading contexts, MEV protection may also be a relevant consideration. If trade orders can be front-run prior to settlement, on-chain confidentiality alone may be insufficient.

Vendor	Trust	Biz Logic	Trade	Reg	Demo
COTI	GC	 Available	 Available	 Available	Privex
Linea	ZK	 Proto	 Proto	 Proto	—
Nightfall	ZK	 Partial	 Early Access	 Available	GitHub
Paladin	ZK/Notary	 Available	 Available	 Available	Docs
Prividium	ZK	 Available	 Available	 Available	Video
Polygon	ZK	 Available	 Available	 Available	Docs
Silent Data	TEE	 Available	 Available	 Available	GitHub

The privacy landscape is evolving very fast. This maturity matrix provides a snap shot at March 2026. This report will be updated every 6 months and we urge the reader to take this into consideration. What you see right now can be very different in a couple of months.

Solution 1: COTI

Summary: COTI is the programmable privacy layer for Web3. Powered by high-performance Garbled Circuits, it delivers fast, low-cost, flexible, and compliant privacy across blockchains, and natively on its Ethereum Layer 2. COTI's architecture enables enterprises and developers to build privacy-preserving applications with end-to-end confidentiality and cost-efficiency at scale. It unlocks a wide variety of use cases in private DeFi, tokenized real-world assets (RWAs), trading, payments, and decentralized identity. COTI leads in private DeFi and enterprise adoption, with active collaborations in banking, government, and global forums.

Privacy Approach

COTI uses Garbled Circuits (GC), an advanced multi-party computation (MPC) technique, to enable encrypted computation within workflows. Sensitive data including inputs, balances, on-chain state, and smart contract logic, remains encrypted end-to-end across storage, computation and transmission. "Input Text" is converted into "Garbled Text", where encrypted Boolean gates are securely evaluated, supporting complex operations over private data without exposing data. Garbled Circuits deliver strong confidentiality with high throughput, low latency, low cost, EVM compatibility, and multi-chain support. Programmable smart contracts allow selective disclosure for audits, compliance, or permissions, keeping data private yet revealable to authorized parties via permissioned view-keys.

Integration Requirements

- Deploy on COTI network or bridge existing assets
- Standard Solidity with parameters specifying confidential data elements
- TypeScript, Python, Hardhat tooling available

Case Studies

- [Central Bank of Europe - Digital Euro](#); Pioneer Partner with the European Central Bank to establish confidential consumer applications on top of a Euro CBDC settlement layer.
- [Privex](#); A COTI native privacy-preserving perpetual exchange that processes in excess of \$25bn in aggregate volume and extends both confidential user and agentic trading.
- [Unicef, Bangladesh Government, StaTwig](#); Integrated COTI for a national-scale rollout in Bangladesh, completing 10 million private computations on encrypted IoT-enabled vaccine data for approximately 7 USD total.

Solution Characteristics (as stated by providers)

- Extends confidential tokens, concealing transaction values and account balances
- Real-time, scalable, confidential computations; private transactions at \$0.0000007
- Multi-party: Extends to scenarios where multiple parties need to contribute private data
- Performative: 1000x faster & 250x lighter than alternative privacy solutions (FHE)
- Roadmap: Multichain expansion, extending a privacy-on-demand capability that can be used on 70+ chains (planned 2026)
- Maturity: Production ready

Github Libraries & Developer Documentation

- [Documentation](#)
- [GitHub](#)

Solution 2: Linea Enterprise (Consensys)

Summary: An Ethereum Layer 2 solution that can be deployed as a private, permissioned "Validium" where the state remains private. Zero knowledge proofs are used to prove the validity of the state while keeping transaction data confidential as well as to enable trustless interoperability with Ethereum. This guarantees full access to the Ethereum ecosystem including its liquidity while providing institutional privacy.

Privacy Approach

Uses a **Private Validium** architecture combined with an Access Control Engine. Transaction data and ledger state stay within the operator-controlled infrastructure and are never published externally. Only zero-knowledge validity proofs (ZKPs) and commitments to these data are posted to Ethereum to prove correctness. Role-Based Access Control (RBAC) restricts data visibility so that participants (auditors, banks) only see the specific transaction data they are entitled to view.

Integration Requirements

- Existing Smart Contracts: Full EVM equivalence allows existing Ethereum smart contracts to deploy without modification.
- Tokens: Native support for all ERC standards (ERC-20, ERC-721, ERC-1155, ERC-3643).
- Tooling: Compatible with standard Ethereum developer tools (Hardhat, Foundry).
- Identity: Users/Institutions can integrate with the Linea Enterprise Gateway (e.g., via standards like OAuth2, wallet signing).

Case Studies

- SWIFT September 2025 pilot: 12+ banks including BNP Paribas, BNY Mellon
- SharpLink Gaming: \$170M ETH with Anchorage Digital custody
- Total value secure on Linea Mainnet public chain: \$750M+ per public sources

Solution Characteristics (as stated by providers)

- Throughput: benchmarks with ~1000 TPS for simple transfers, and a roadmap to 30,000+ TPS.
- Finality: instant/single-slot finality on Layer 2; L1 finality depends on proof posting frequency.
- Cost: sub-\$0.01 per transaction.
- Maturity: production ready, the core stack has been running Linea mainnet for two years.

Github Libraries & Developer Documentation

Github: <https://github.com/Consensys/linea-monorepo>

Documentation: <https://docs.linea.build/stack>

Solution 3: Nightfall (EY)

Summary: Nightfall_4 is an open-source zero-knowledge-based privacy protocol developed and maintained by EY that enables confidential token transfers on Ethereum-compatible networks. It uses cryptographic commitments and zero-knowledge validity proofs to ensure transaction correctness while keeping amounts and counterparties private. The protocol is designed for enterprise and institutional use cases, supporting selective disclosure for compliance and audit without introducing global decryption keys or protocol backdoors. Identity, permissioning, and KYC integration are deployment-configurable rather than hard-coded.

Nightfall_4 focuses on practical privacy for regulated environments, balancing confidentiality, auditability, and interoperability with public blockchain infrastructure

Privacy Approach

Converts tokens (ERC-20, ERC-721, ERC-1155, ERC-3525) into cryptographic 'commitments' using ZK-ZK rollup architecture. Validity proofs verify correctness without revealing amounts, parties, or transfer details.

Integration Requirements

- Tokens deposit to receive private commitments
- Supports decentralized permissioning and KYC-gating
- X.509 enterprise identity certificates supported for deposit/withdrawal
- Gas fee abstraction available for ERC-20 tokens

Case Studies

- Starknet Ecosystem: Integration to enable high-throughput privacy-preserving payments leveraging zk-based scalability infrastructure.
- Plume Network: Layer-3 deployment for privacy-enabled institutional RWA tokenization with sequencer-level AML compliance.
- Celo Foundation: First payments-focused blockchain to deploy Nightfall; targeting enterprise B2B cross-border payment(\$180T market)
- Microsoft: Supply chain proof-of-concept

Solution Characteristics (as stated by providers)

- Cost: 3.5x cost reduction vs. public ERC-20 transfers
- Development history: Since 2019
- Open source: Fully public domain, no commercial licensing required
- Business logic: Smart contract code remains publicly verifiable to preserve transparency and composability, while transaction amounts, counterparties, and token balances are cryptographically shielded from public view
- Maturity: Deployed across multiple networks including Plume Network, Celo Foundation, and Starknet Ecosystem

Github Libraries & Developer Documentation

GitHub: https://github.com/EYBlockchain/nightfall_4_CE

Solution 4: Paladin (LFDT, Kaleido)

Summary: Paladin, a top-level project of the Linux Foundation Decentralized Trust (LFDT), is a unified framework for "Programmable Privacy" designed to bring enterprise-grade confidentiality to any Ethereum Virtual Machine (EVM) compatible blockchain. Paladin is a modular runtime that allows developers to plug in diverse cryptographic technologies such as Zero-Knowledge Proofs (ZKP), private notary systems, and Fully Homomorphic Encryption (FHE). Its standout feature is its ability to facilitate atomic, all-or-nothing transactions across multiple privacy domains, enabling complex business logic (e.g., Delivery vs. Payment) where different legs of a trade might use different privacy models (like a ZKP-based token vs. a notary-based system).

Privacy Approach

Paladin achieves confidentiality via localized **Privacy Domains**, with three built-in protocols:

- Pente, private smart contract execution through "ephemeral EVMS" with state commitments recorded on-chain
- Zeto, a ZKP based token implementation
- Noto, a notary based token implementation for regulated assets

By abstracting diverse cryptographic methods, including ZKP, FHE, and off-chain notarization, into a single modular runtime, Paladin allows enterprises to orchestrate **atomic transactions** that bridge multiple privacy models.

Integration Requirements

- Deployment: each participant hosts a Paladin Node that runs as a sidecar to any standard EVM client.
- Privacy Domain: admins defines and deploys specific "Domains", such as Zeto, Noto or Pente, which dictate the cryptographic rules and verifier contracts on the base ledger.
- Identity & Key Management: integration with enterprise-grade KMS or HSM is built-in.
- Peer-to-Peer Connectivity: nodes communicate via a secure connection layer, typically via mTLS, for off-chain data exchange and multi-party signature gathering.
- Off-chain State Store: each node uses a dedicated database for the private states.

Case Studies

- Central Bank of Brazil, Drex Project - Mastercard, Banco BV phase 2 use cases
- [Bank Indonesia, Digital Rupiah](#) - Cash Ledger
- [Hong Kong Monetary Authority, eHKD Phase 2](#) - Zeto was tested as a Private Enhancing Technology

Solution Characteristics (as stated by providers)

- Programmable Privacy & Atomic Interoperability
- High-Performance UTXO-based out-of-box privacy tokens
- Pluggable Cryptography
- Maturity: adopted by many influential projects in the financial industry

Github Libraries & Developer Documentation

Source code: <https://github.com/LFDT-Paladin/paladin>

Documentation: <https://lfdt-paladin.github.io/paladin>

Solution 5: Prividium (ZKsync / Matter Labs)

Summary: Enterprise-grade private, permissioned Layer 2 infrastructure that combines institutional privacy controls with native Ethereum connectivity and Prividium-to-Prividium interoperability. Transactions execute privately within enterprise-controlled environments while ZK proofs anchor integrity to Ethereum—enabling compliant tokenization with access to public liquidity when needed. Prividiums can be deployed as multiple networks that remain private and isolated but with native interoperability to enable multi parties to transact with each other privately while maintaining full control of their own environment.

Privacy Approach

Permissioned ZKsync Layer 2 architecture with zero-knowledge proofs. Transaction data executes off-chain within operator-controlled infrastructure. Only ZK validity proofs and state root commitments post to Ethereum. No transaction details, amounts, or counterparty information ever reach outside the Prividium. Role-based access controls define granular permissions at the smart contract function level. Selective disclosure enables audit access via Merkle proofs without exposing data to unauthorized parties.

Integration Requirements

- Full EVM equivalence; existing Solidity contracts deploy without modification with standard tooling supported (Hardhat, Truffle, Foundry)
- Deployment options: Matter Labs managed services, customer cloud, or fully on-premises
- Enterprise SSO integration (Okta, Azure AD, etc)
- Assets bridge to Prividium environment with native interoperability to other ZKsync chains and Ethereum. Assets can remain on Ethereum L1 where they can be interacted with through ZKsync's Layer 1 Interop
- Commercial agreement with Matter Labs required

Case Studies

- [Cari Network: Tokenized deposits network for regional and community banks in the US. Currently on testnet](#)
- [Memento Blockchain with Deutsche Bank Project DAMA 2: Tokenized fund infrastructure currently in production](#)
- [35+ Global Financial Institutions: Joined interoperable Prividium pilot](#)

Solution Characteristics (as stated by providers)

- Cost: Sub-\$.001 per transaction for entire system including Layer 1 settlement fees with proving costs of \$.00013 per transaction
- Performance: 15,000+ transactions per second supported per Prividium. ZK proof generation time is sub-1 second. 200ms blocktimes with 100ms achievable
- Ecosystem: Cross-chain privacy limited to ZKsync ecosystem with direct Layer 1 interoperability
- Licensing: Commercial relationship with Matter Labs required
- Maturity: In early production state with multiple co-design partners

Github Libraries & Developer Documentation

- Prividium [NPM package](#)
- Local-prividium [GitHub repo](#)

- [Public docs](#)
- [Public product page](#)

Solution 6: Polygon CDK

Summary: A modular, open-source multistack toolkit by Polygon Labs for launching sovereign, ZK-secured EVM chains with configurable, financial-grade privacy, enterprise-grade controls, and native connectivity to Agglayer for access to Ethereum liquidity.

Privacy Approach Privacy with Polygon CDK is a configurable spectrum, rather than a binary on/off switch. Institutions design the right combination of controls for a specific regulatory environment, counterparty structure, and use case. Available privacy configurations include:

- Private data availability (Validium mode): Transaction data is stored off-chain in operator-controlled infrastructure, rather than posted to Ethereum L1. This ensures the underlying data layer is not exposed beyond authorized parties, while ZK validity proofs still guarantee correctness on-chain.
- Private RPC, mempool, and block explorers (via Gateway): Granular privacy controls with IAM permissions geared to each user. Users only see transactions relevant to them, not the full chain state.
- ZK Validity proofs: Only cryptographic commitments are submitted to the chain, not raw transaction data. Counterparties and regulators can verify correctness without seeing underlying transaction details.
- Access control lists (ACLs): Role-based permissions define what each participant can do. For sensitive transactions, amounts and counterparties can be hidden from node operators while remaining selectively revealable for an audit.
- Identity and compliance tooling: KYC out-of-the-box via Billions identity; integrations with Microsoft Entra and AWS IAM for enterprise SSO.

Fully homomorphic encryption (FHE) via Zama: For use cases requiring computation on encrypted data without ever decrypting it. Currently in implementation.

Integration Requirements

- Full EVM compatibility; existing smart contracts deploy without modification
- Direct upgrade path from Hyperledger Besu for institutions with existing permissioned networks
- Multiclient options, including Erigon client (stress-tested, audited), CDK op-geth (fork of OP Stack), CDK op-geth Validium (forthcoming with Succinct and Conduit), and CDK op-reth (forthcoming)
- Options: zkRollup, Validium, or Sovereign modes

Case Studies

- Polygon CDK launched an enterprise-focused, privacy-first rollup in Q3 2025, [per Messari](#)
- Polygon PoS network: \$2.4T stablecoin volume, \$1.2B locked, 164M+ unique addresses, 6.7B+ transactions processed per vendor
- CDK adopters include: Katana, OKX, Immutable, Silicon, Lumia, and others, per vendor

Solution Characteristics (as stated by providers)

- Privacy model: Configurable spectrum, including Validium (off-chain data storage with ZK proofs to Ethereum), private block explorers, private RPC/mempool, ACL-based role permissions, FHE for computation of encrypted data, and selective disclosure for regulators/auditors
- Migration: Vendor describes direct upgrade path from Hyperledger Besu, with two proven paths (fast snapshot, full-history)
- Roadmap: Performance is currently up to 60+ MGas/s burst throughput, with increased capacity and diversity of clients in upcoming versions
- Maturity: Vendor describes CDK Erigon as production ready (Q3 2025 launch); CDK op-geth Validium in production and forthcoming public release; CDK op-reth forthcoming

Github Libraries & Developer Documentation

Polygon CDK documentation: <https://docs.agglayer.dev/cdk/get-started/overview/>

Github: <https://github.com/0xPolygon/cdk-erigon>

Solution 7: Silent Data (Applied Blockchain)

Summary: Silent Data is an Ethereum Layer 2 developed by Applied Blockchain that provides full Turing-complete programmable privacy for smart contract execution and transaction data. Privacy is enforced at the execution layer using trusted execution environments (TEEs), enabling regulated and enterprise applications to control what information is publicly disclosed while maintaining standard EVM execution and connectivity to Ethereum. As well as having a public Layer 2, Silent Data can be deployed as a dedicated stack for organizations that want their own privacy enabled L2 with connection to the Ethereum ecosystem.

Privacy Approach

Uses Trusted Execution Environments (TEEs)—specialized chips that process data in encrypted form. Infrastructure operators cannot observe processing. Privacy guarantees derive from hardware properties. The result is a Turing-complete, high performance EVM privacy solution. Privacy controls are applied at the execution layer, allowing transaction inputs, intermediate state, and smart contract logic to execute confidentially. No custom cryptography. No changes to Solidity or standard blockchain code. Everything just works as-is, and everything in Solidity can be made private.

Integration Requirements

- Existing smart contracts deploy directly (standard EVM, no custom extensions, no bespoke cryptography, no modification to open source blockchain software)
- First L2 deployed using OP Stack, part of Ethereum Superchain ecosystem
- Assets bridge to Silent Data Ethereum L2 network (similar to Base, Unichain etc, same stack)
- Works with standard Ethereum / EVM tooling (wallets, development frameworks, dapps).

Case Studies

- [Archax](#): FCA-regulated digital securities exchange deployed Aberdeen money market fund (MMF) live in production.

- [DHL Health Logistics – CRYOPDP](#): Award-winning industry supply chain logistics platform. Operates in production, supporting privacy-preserving data sharing across regulated healthcare and life sciences logistics workflows.
- Shell: Energy trading applications with privacy.
- [Blueprints](#): Pre-built, privacy-enabled application templates, including supply chain data sharing, asset tokenization, and decentralized exchange (DEX) use cases.

Solution Characteristics (as stated by providers)

- Performance: Up to 4,000 transactions per second (uniquely blockchain performance not impacted by addition of privacy protocol)
- Trust model: Privacy guarantees depend on hardware manufacturer attestation (Intel)
- Network: Standard Ethereum L2, assets can be bridged
- Maturity: Silent Data L2 is live on OP L2 Superchain (verified by OP and L2Beat)
- Business logic: Designed to support complex enterprise workflows, including regulated trading, asset issuance, settlement, and supply-chain data sharing

Github Libraries & Developer Documentation

- [Documentation](#)
- [GitHub](#)

Decision Framework

Questions to support internal evaluation.

Privacy Requirements

- *What specific information must remain confidential?*
- *Who must be able to see this information?*
- *Under what conditions should information be disclosed?*
- *What are the consequences if this information becomes public?*

Current Architecture

- *What token standards are currently deployed?*
- *Can existing smart contracts be modified?*
- *What custody infrastructure is in place?*
- *What counterparties must be involved, and can they adopt new systems?*

Integration Constraints

- *What is the acceptable timeline for implementation?*
- *What internal resources are available?*
- *Are there requirements for open-source licensing?*
- *What are the requirements for vendor support and SLAs?*

Trust Model

- *What trust assumptions are acceptable? (hardware-based, mathematical proof-based, trusted third-party)*
- *What level of vendor dependency is acceptable?*

Maturity

- *What level of production deployment history is required?*
- *What independent security validation is required?*

Operations

- *What transaction throughput is required?*
- *What latency and finality requirements exist?*
- *What cost constraints apply?*

Future Outlook & Open Questions

Toward a Standardized Privacy Layer

As we look toward the 2026 - 2030 horizon, the convergence of these technologies suggests the emergence of a hybrid privacy layer for Ethereum. However, several open questions remain:

How will regulatory bodies view "programmable privacy" versus "total anonymity"?

Can we achieve cross-chain privacy standards to prevent liquidity fragmentation?

Joint Statement: The members of this working group, in collaboration with the wider Ethereum community and the Ethereum Foundation (EF), remain committed to the development of "Compliant Privacy." We believe that the future of the enterprise web is not just decentralized, but inherently confidential.

Additional Privacy Solutions

Solutions profiled were contributed by EEA member organizations participating in the Privacy Working Group. Inclusion requires EEA membership and active working group participation. Other privacy solutions exist in the Ethereum ecosystem. These are listed below based on publicly available information. Organizations interested in contributing detailed profiles may contact the EEA about membership.

Compagnies	Solution
RAILGUN DAO	Privacy system enabling private transfers and DeFi interactions on Ethereum mainnet using ZK proofs
Aztec Labs	ZK rollup with native private smart contracts and encrypted state on Ethereum
Zama	Encrypted smart contracts using fully homomorphic encryption on EVM chains
Fhenix	FHE-powered L2 enabling computation on encrypted data
Panther	Privacy pools for compliant private transactions with selective disclosure
ScopeLift /Umbra	Stealth address protocol for private payments on Ethereum

Glossary & Appendices

Glossary

To be written post report completion

Contact

To join the Enterprise Ethereum Alliance contact membership@entethalliance.org

Organization	Solution	Representatives / Role	Contact
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Enterprise Ethereum Alliance